Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

If P be the point $(1, 0)$ and Q, a point on the locus $y^2 = 8x$. The locus of the mid point of PQ is:

(A) $x^2 + 4y + 2 = 0$

(B) $x^2 - 4y + 2 = 0$

(C) $y^2 - 4x + 2 = 0$

(D) $y^2 + 4x + 2 = 0$

Q.2

The locus of the mid-point of the line segment joining the focus to a moving point on the parabola $y^2 = 4ax$ is another parabola with directrix

(A) $x = -a$

(B) $x = -a/2$

(C) $x = 0$

(D) $x = a/2$

Q.3

The locus of the vertices of the family of parabolas $y = \frac{a^3x^2}{3} + \frac{a^2x}{2} - 2a$ is

(A) $xy = 64/105$

(B) $xy = 105/64$

(C) $xy = 3/4$

(D) $xy = 35/16$

Q.4

If two tangents drawn from a point P to the parabola $y^2 = 4x$ are at right angles, then the locus of P is

(A) $x = -1$

(B) $2x - 1 = 0$

(C) $x = 1$

(D) $2x + 1 = 0$

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Q.5

Let O be the vertex and Q be any point on the parabola $x^2 = 8y$. If the point P divides the line segment OQ internally in the ratio $1:3$, then the locus of P is:

(A) $x^2 = y$

(B) $y^2 = x$

(C) $y^2 = 2x$

(D) $x^2 = 2y$

Q.6

Let (x, y) be any point on the parabola $y^2 = 4x$. Let P be the point that divides the line segment from $(0, 0)$ to (x, y) in the ratio $2:3$. Then locus of P is

(A) $x^2 = y$

(B) $5y^2 = 2x$

(C) $5y^2 = 8x$

(D) $5x^2 = 2y$

Q.7

PQ is a double ordinate of a parabola $y^2 = 4ax$. The locus of its points of trisection is another parabola length of whose latus rectum is k times the length of the latus rectum of the given parabola, the value of k is

(A) $1/9$

(B) $1/3$

(C) $2/3$

(D) none of these

Q.8

The centre C of a variable circle passing through a fixed point $(a, 0), a > 0$, touches the line $y = x$. Locus of C is a parabola whose directrix is

(A) $x + y = 0$

(B) $x + y = a$

(C) $x - y = 0$

(D) $x - y = a$

Q.9

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If P is the point $(1,0)$ and Q lies on the parabola $y^2 = 36x$, then the locus of the mid point of PQ is:

(A) $y^2 = 9(2x-1)$

(B) $y^2 = 9(x+2)$

(C) $y^2 = 2(x-9)$

(D) $x^2 = 9(y-2)$

Q.10

Locus of the mid points of the chords of the parabola $y^2 = 8x$ which touch the circle $x^2 + y^2 = 4$ is

(A) $\left(y^2 - 4x\right)^2 = 16\left(x^2 + 4\right)$

(B) $\left(y^2 - 4x^2\right) = 4\left(x^2 + a^2\right)$

(C) $\left(y^2 - 4x\right)^2 = 4\left(16 + y^2\right)$

(D) $\left(y^2 - 4x\right)^2 = 16\left(4 + y^2\right)$

Q.11

The tangent PT and the normal PN to the parabola $y^2 = 4ax$ at a point P on it meet axis at points T and N, respectively. The locus of the centroid of the triangle PTN is a parabola whose vertex is the point:

(A) $\left(\frac{a}{3}, 0\right)$

(B) $\left(\frac{2a}{3}, 0\right)$

(C) $(a, 0)$

(D) $\left(\frac{4a}{3}, 0\right)$

Q.12

A line PQ meets the parabola $\mathrm{y^2 = 4ax}$ in R such that PQ is bisected at R. If the coordinates of P are $\mathrm{(x_1, y_1)}$ then the locus of Q is the parabola

(A) $\left(y + y_1\right)^2 = 8a\left(x + x_1\right)$

(B) $\left(y - y_1\right)^2 = 8a\left(x + x_1\right)$

(C) $\left(y + y_1\right)^2 = 8a\left(x - x_1\right)$

(D) None of these

Q.13

A point moves such that the square of its distance from a straight line is equal to the difference between the square of its distance from the centre of a circle and the square of the radius of the circle. The locus of the point is

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(A) a straight line at right angle to the given line

(B) a circle concentric with the given circle

(C) a parabola with its axis parallel to the given line

(D) a parabola with its axis perpendicular to the given line

Q.14

Equation of the parabola whose vertex is $(-3, -2)$, axis is horizontal and which passes through the point $(1, 2)$ is

(A) $y^2 + 4y + 4x - 8 = 0$

(B) $y^2 + 4y - 4x + 8 = 0$

(C) $y^2 + 4y - 4x - 8 = 0$

(D) None of these

Q.15

Let P be the point $(1, 0)$ and Q a point on the locus $y^2 = 8x$. The locus of mid point of PQ is

(A) $x^2 + 4y + 2 = 0$

(B) $x^2 - 4y + 2 = 0$

(C) $y^2 - 4x + 2 = 0$

(D) $y^2 + 4x + 2 = 0$

Answers & Solutions

Q.1 Answer:

$y^2 - 4x + 2 = 0$

Solution:

Let the coordinates of Q be $\left(2t^2, 4t\right)$
coordinates of the mid point of PQ are $\left(\frac{2t^2+1}{2}, 2t\right)$
Let $h = \frac{2t^2+1}{2}, k = 2t$
Eliminating t, we get $4h = k^2 + 2$ So the required locus is $y^2 - 4x + 2 = 0$

Q.2 Answer:

$x = 0$

Solution:

The focus of the parabola $y^2 = 4ax$ is $S(a, 0)$, let $P\left(at^2, 2at\right)$ be any point on the parabola then coordinates of the mid-point of SP are given by

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$x = \frac{a(t^2+1)}{2}, \quad y = \frac{2at+0}{2}$

Eliminating 't' we get the locus of the mid-point as $y^2 = 2ax - a^2$ or $y^2 = 2a(x - a/2)$ (1) which is a parabola of the form $Y^2 = 4AX$ (2)

Where $Y = y, X = x - a/2$ and $A = a/2$

Equation of the directrix of (2) is $X = -A$ So equation the directrix of (1) is $x - a/2 = -a/2$

$\Rightarrow x = 0$

Q.3 Answer:

$xy = 105/64$

Solution:

Equation of the parabola can be written as

$$\frac{y}{a} = \left(\frac{ax}{\sqrt{3}} + \frac{\sqrt{3}}{4}\right)^2 - \frac{3}{16} - 2$$

$$\text{Or } \left(x + \frac{3}{4a}\right)^2 = \frac{a^2}{3a}\left(y + \frac{35}{16}a\right)$$

vertex is $x = -3/4a, y = -35a/16$

Locus of the vertex is $xy = 105/64$.

Q.4 Answer:

$x = -1$

Solution:

Equation of a tangent to the parabola is $y = mx + 1/m$.

If it passes through $P(h, k)$, then $k = mh + 1/m$

$\Rightarrow m^2h - mk + 1 = 0$ **which gives the slopes of two tangents passing through P.**

As these are at right angles, product of the slopes $= -1 \Rightarrow 1/h = -1$

Locus of P is $x = -1$ which is the directrix of the parabola.

Q.5 Answer:

$x^2 = 2y$

Solution:

Let the coordinate of Q be (x', y') and that of P be (h, k), then

$h = \frac{1}{4}x', k = \frac{1}{4}y'$

As (x', y') lies on the parabola $x^2 = 8y$ we get $(4h)^2 = 8(4k)$

$\Rightarrow$

$h^2 = 2k$

Locus of $P(h, k)$ is $x^2 = 2y$

Q.6 Answer:

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$5y^2 = 8x$

Solution:

Let P be (h, k) then $h = \frac{2x}{5}, \quad k = \frac{2y}{5}$

As (x, y) lies on the parabola $y^2 = 4x$.

$\left(\frac{5k}{2}\right)^2 = 4\left(\frac{5h}{2}\right)$

$\Rightarrow 5k^2 = 8h$

$\Rightarrow$ Locus of P is $5y^2 = 8x$.

Q.7 Answer:

$1/9$

Solution:

Let PQ be a double ordinate of the parabola $y^2 = 4ax$ meeting the axis at L, R and S be the points of trisection of PQ.

So that $PR = RS = SQ$. (Fig.)

If the coordinates of R are (h, k), the coordinates of S are $(h, -k)$

the coordinates of P are $(h, 3k)$ Since P lies on the parabola $y^2 = 4ax$

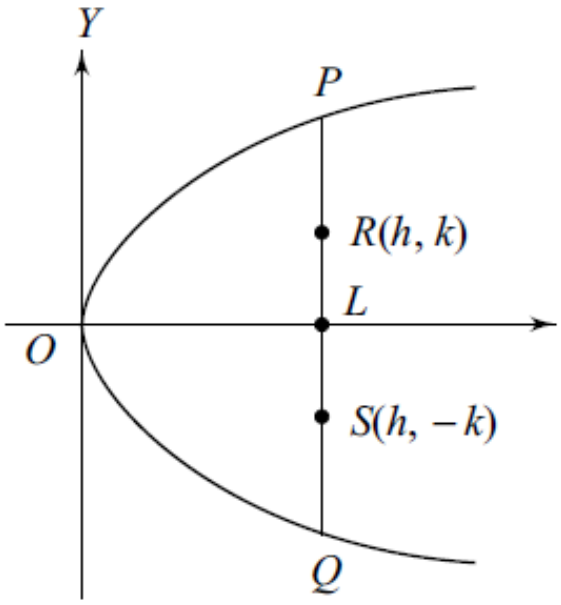
$(3k)^2 = 4ah \quad \Rightarrow \quad 9k^2 = 4ah$

The locus of $R(h, k)$ is therefore

$9y^2 = 4ax$ or $y^2 = (4/9)ax$

which is a parabola whose length of the latus rectum $= (4a/9)$

$= (1/9)$ (length of latus rectum of the given parabola)



Q.8 Answer:

$x - y = 0$

Solution:

C is equidistant from the point $(a, 0)$ and the line $y = x$, so locus of C is a parabola with directrix $y = x$ i.e. $x - y = 0$

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Q.9 Answer:

$y^2 = 9(2x - 1)$

Solution:

Let the coordinates of Q be $\left(9t^2, 18t\right)$ coordinates of R the mid point of PQ are $\left(\frac{9t^2+1}{2}, 9t\right) = (x, y)$. Locus of R is

$2x = 9\left(\frac{y}{9}\right)^2 + 1$

$\Rightarrow y^2 = 9(2x - 1)$

Q.10 Answer:

$\left(y^2 - 4x\right)^2 = 4\left(16 + y^2\right)$

Solution:

Let (h, k) be the mid-point of the chord of the parabola $y^2 = 8x$, then its equation is

$ky - 4(x + h) = k^2 - 8h$

$\Rightarrow 4x - ky + k^2 - 4h = 0$

which touches the circle $x^2 + y^2 = 4$

$\Rightarrow \frac{k^2-4h}{\sqrt{16+k^2}} = 2$

$\Rightarrow \left(k^2 - 4h\right)^2 = 4\left(k^2 + 16\right)$

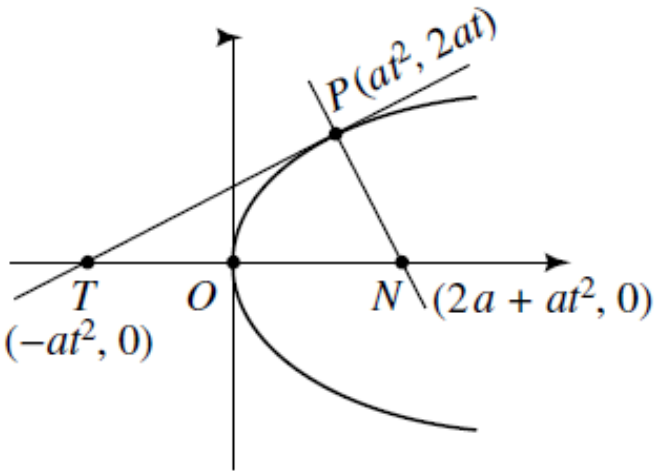
Locus of (h, k) is $\left(y^2 - 4x\right)^2 = 4\left(16 + y^2\right)$

Q.11 Answer:

$\left(\frac{2a}{3}, 0\right)$

Solution:

Equations of tangent and normal at point $P\left(at^2, 2at\right)$ to the parabola are $ty = x + at^2$ $y = -tx + 2at + at^3$ respectively.



These meet the axis of the parabola at $T\left(-at^2, 0\right)$ and $N\left(2a + at^2, 0\right)$

Let centroid of ΔPTN be (h, k), then $h = \frac{1}{3}\left(-at^2 + 2a + at^2 + at^2\right) = \frac{1}{3}\left(2a + at^2\right)$ and $k = \frac{1}{3}(0 + 0 + 2at) = \frac{2}{3}(at)$

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Eliminating t, we get

$\left(h-\frac{2a}{3}\right)=\frac{a}{3}\left(\frac{3k}{2a}\right)^2$

Thus, locus of centroid is

$y^2=\frac{4}{3}a\left(x-\frac{2a}{3}\right)$

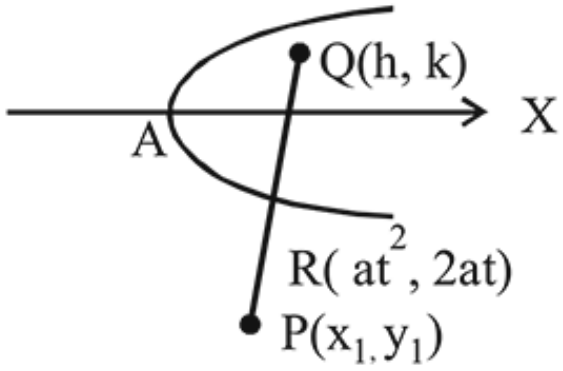
Vertex of this parabola is $(2a/3,0)$;

Q.12 Answer:

$(y+y_1)^2=8a\,(x+x_1)$

Solution:

Let the coordinates of Q **be** (h,k)**. Since the point** R **lies on the parabola. let its coordinates be** $(at^2,2at)$



Since R **is mid point of** PQ**,**

$\therefore at^2=\frac{x_1+h}{2}$ **and** $2at=\frac{y_1+k}{2}$

$\Rightarrow \quad t^2=\frac{x_1+h}{2a}$ **and** $t=\frac{y_1+k}{4a}$

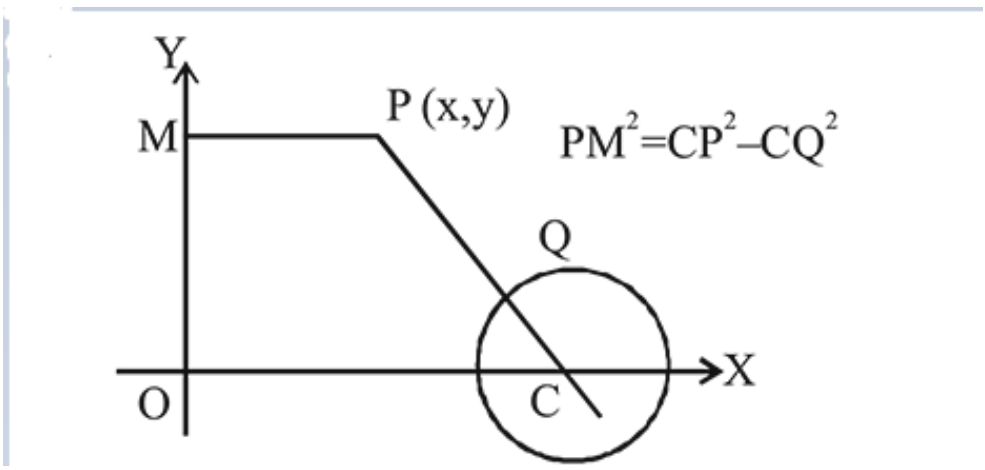
Equating the two values of t, we get $\left(\frac{y_1+k}{4a}\right)^2=\frac{x_1+h}{2a}\Rightarrow(y_1+k)^2=8a\,(x_1+h)$

Hence, locus of $\mathrm{Q}(\mathrm{h},\mathrm{k})$ **is** $(y+y_1)^2=8a\,(x+x_1)$

Q.13 Answer:

a parabola with its axis perpendicular to the given line

Solution:



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Let the given line be the y -axis and the circle to have the eq. $x^2+y^2+2gx+2fy+c=0$ then according to given condition $x^2=(x+g)^2+(y+f)^2-\left(g^2+f^2-c\right)$
$\Rightarrow (y+f)^2=-2g\left(x-\frac{f^2-c}{2g}\right)$
which represents a parabola with its axis $\perp$ toy-axis.

Q.14 Answer:

$y^2+4y-4x-8=0$

Solution:

Since the axis is horizontal and vertex is $(-3,-2)$
$\therefore$ the equation of the parabola must be of the form $(y+2)^2=4a(x+3)$
It passes through $(1,2)$, so $16=16$ a i.e $\mathrm{a}=1$. Hence, the equation of the required parabola is $(y+2)^2=4(x+3)$ or $y^2+4y-4x-8=0$

Q.15 Answer:

$y^2-4x+2=0$

Solution:

Lat the point Q be $\left(2t^2,4t\right)$ and mid point of PQ be (h,k) then $h=\frac{1+2t^2}{2}$ and $k=\frac{0+4t}{2}$
Eliminating t, we get $y^2-4x+2=0$.

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